



Improving properties of soy protein–based hydrogel composites by incorporating bamboo biochar towards slow release and water retention of fertilizers and enhanced plant growth

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Abstract

The development of environmentally friendly slow-release fertilizers with effective water retention is an urgent need in modern agriculture. Although biochar can improve soil fertility as a nutrient carrier, it suffers from poor slow-release performance and water retention. Conversely, soy protein hydrogels, characterized by their hydrophilic nature with three-dimensional cross-linked networks, can retain large amounts of water and facilitate the slow release of fertilizers and water due to their high specific surface area. Hence, a novel slow-release composite material with high water retention was prepared by introducing bamboo biochar into a soy protein–based hydrogel (SPB) network through graft copolymerization. The findings indicated that the bamboo biochar promoted the SPB cross-linked network density, which improved the swelling rate of SPB materials and soil water-holding capacity. Moreover, SPB-2–4% and SPB-3–4% exhibited superior slow-release capabilities for nitrogen fertilizer. Cucumber seedlings treated with SPB materials containing bamboo biochar demonstrated enhanced growth and chlorophyll content than those treated with biochar-free SPB materials. Compared with the control, the cucumber plants treated with SPB-2–4% displayed a significant increase in fresh weight, root length, and leaf area by 139.32%, 99.20%, and 149.45%, respectively, which can be attributed to the positive synergistic effect of soy protein and bamboo biochar. Furthermore, the nutrients and porous structures of bamboo biochar favor the proliferation of microorganisms, enriching the soil microbial community. Therefore, the bamboo biochar-soybean protein hydrogel composites have great application prospects for sustainable agriculture and provide a new direction for the development of slow-release and water-retention fertilizers.

Keywords Bamboo biochar · Soy protein · Hydrogel · Water retention · Slow-release · Plant growth

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1 Introduction

For many years, agroforestry has played a pivotal role in the development of the world economy. With socio-economic development and the increasing global population, the need for increased food production and urban landscaping is becoming more urgent. Improving crop yield and increasing green spaces in limited planting areas are significant social development problems that need immediate attention. Water is fundamental to plant growth, influencing their morphology, physiological activities, and ecological functions. Additionally, fertilizers provide the necessary nutrients for plant growth, which promote plant growth and crop yield. Thus, water and fertilizer play essential roles in modern agriculture and forestry development. However, water scarcity and rising irrigation costs have severely constrained the

development of agroforestry landscapes and dryland economies [1]. According to a survey, 84% of the world's arable land is dryland, and a large amount of this land is underproductive due to a lack of water, so efficient management and utilization of water resources are essential [2]. At the same time, in recent years, as the demand for food has increased, the use of chemical fertilizers has also increased year by year, but the water solubility and volatility of traditional fertilizers have led to their extremely low utilization rates [3]. The massive use of chemical fertilizers has increased farmers' input costs and also contributed to the eutrophication of water bodies. For instance, nitrogen fertilizer, vital for plant growth, has a utilization rate of only 30–50% [4, 5], whereas urea, a conventional nitrogen fertilizer, is even lower at just 30% [6]. Therefore, there is an urgent need to develop a water-fertilizer-integrated slow-release fertilizer to address the problems confronting modern agricultural development.

In recent years, the application of biochar in slow-release fertilizers has shown great potential. Biochar is a by-product obtained by burning biomass under limited oxygen conditions [7]. Biochar has rich functional groups on its surface and a porous carbon structure internally, which plays critical roles in soil pollution control, nutrient retention, and soil ecological regulation [8–11]; hence, it is a promising soil improvement material [12]. He et al. showed that biochar can be used as an adsorbent to adsorb ammonium and phosphate from wastewater and could release these nutrients steadily for a long period to promote the growth of maize seedlings in pot experiments [13]. Marcińczyk's study indicates that co-pyrolysis, co-composting, and co-application of biochar with synthetic fertilizers can effectively improve fertilizer efficiency and promote plant growth [14]. However, the effect of biochar on soil water retention was minimal compared to the quantity added. Amoakwah et al. revealed little or no significant impact on soil water retention when corn cob biochar was applied at 10 t/hm², whereas an obvious significance was only observed when 20 t/hm² was applied. Furthermore, both Randolph et al. and Sun et al. showed that biochar was insufficient to dilute the mineral fraction of the soil and that its porous nature and lower density led to a reduction in soil bulk density [15–17]. Therefore, improving the water retention capacity of the material is a major issue facing biochar-based slow-release fertilizers.

Based on the demand for water retention capacity, hydrogels with three-dimensional cross-linked networks are favored as hydrophilic polymers that can absorb and retain large amounts of water [18, 19]. At the same time, their internal porous network and swelling properties equip hydrogels with the potential to release nutrients sustainably [20]. Most of the current hydrogels are obtained through the free radical polymerization of polymers such as acrylic acid and acrylamide. The high water absorption of hydrogels is due to the large number of hydrophilic groups in

the polymer molecular chain ($-\text{COO}^-$, $-\text{OH}$, and $-\text{NH}_2$) [21]. However, the high cost of acrylamide makes it difficult to popularize the application of hydrogels in agricultural production. Additionally, the slow biodegradation of acrylamide poses a certain threat to the environment and human health [22]. Therefore, more and more scholars are devoted to the research of renewable resources and biodegradable hydrogels. Sarmah et al. prepared a highly absorbent biodegradable hydrogel by combining starch and acrylic acid (AA) in a ratio of 1:1, which could increase soil water-holding capacity by 120% with the addition of 0.25% and effectively improve soil porosity and weight capacity [23]. Song et al. prepared hydrogels by reacting the natural polymers, potassium humate, and sodium alginate, with acrylic acid and loading urea into them. When the composite material was added to the soil at a 0.2% concentration, it significantly promoted plant growth, resulting in a 21.1% decrease in malondialdehyde, a 31.4% increase in chlorophyll, and a 33.2% increase in soluble soybean protein in cabbage leaves [24]. Yang et al. grafted copolymerized corn stover cellulose with acrylic acid and potassium humate to produce a hydrogel with high degradability, exhibiting 32.02% degradation within 50 days. The degraded humic acid can serve as a nutrient to promote plant growth, and its water absorption rate remains at 410 g/g after five cycles [25].

Soy protein, as a green, low-cost, and readily available biomass, offers significant potential for creating novel materials for development and application in the agricultural field. Rosado et al. prepared a biodegradable material with good slow-release properties by mixing soy protein with zinc in a compression tablet, which has the ability to prolong the zinc release cycle and improve plant uptake and assimilation efficiency [26]. Jose et al. thermally induced gel formation by mixing whey and soy proteins in a salt solution. An increase in the concentration of soy protein in the hydrogel has been shown to reduce the roughness inside the gel and create a more homogeneous internal structure. The increase in the porous structure further enhances the water retention of the material. As the mass percentage of soy protein and whey protein increases from 0 to 10%, water retention can increase from 60 to 90% [27]. Kaur et al. studied the degradability of soy protein-polyethylene graft copolymers using the soil burial method. The findings showed that soy protein could increase the total number of bacterial colonies in the soil and promote the degradation of polyethylene materials, with degradation rates reaching up to 72% in the fourth month. Concurrently, plant growth experiments revealed that the degradation of soy protein-polyethylene copolymers was not only harmless to plant growth but actually promoted it [28]. Yang et al. prepared a biodegradable flowerpot using soy protein and PLA as raw materials, whereby the addition of soy protein promotes the degradation of PLA. The pot experiments showed that these materials can promote

the growth of marigold roots. Moreover, the natural nitrogen fertilizer from the degradation of soy protein, along with the small molecules formed from the degradation of PLA, can replace 80% of the synthetic fertilizers needed for plant growth [29]. Numerous studies have shown that new biomass materials prepared from plant proteins have a good potential for agricultural application [30]. However, few studies reported on integrating the biodegradability and slow-release properties of plant protein-based hydrogels for application in plant cultivation.

A novel slow-release composite material was prepared using the solution polymerization method by adding bamboo biochar to a soy protein alkaline solution, followed by incorporating acrylic acid as the reaction monomer and urea as the nitrogen fertilizer, which could be applied in phyto-cultivation. The aim is to integrate the benefits of soy protein-based hydrogels, which act as water retention and slow-release materials and degrade into nitrogen fertilizers, and combine with bamboo biochar which has a porous structure, to obtain a new biomass slow-release fertilizer that is green, low-cost, and simple to prepare [31, 32]. A pot experiment was conducted to verify the effect of the plant protein hydrogel-bamboo biochar slow-release material on plant growth. The prepared soy protein-based hydrogel (SPB) materials exhibited excellent water retention and slow-release properties. In the pot experiment, the SPB material significantly promoted the growth of cucumber, and the addition of bamboo biochar further enhanced the growth-promoting effect of the SPB material on cucumber seedlings. Therefore, the plant protein-biochar composite material, as a new water-retaining and slow-release fertilizer, has great prospects for development in the agroforestry sector.

2 Materials and methods

2.1 Materials

Bamboo wood chips and soy protein (protein content of 95%) were purchased from Anji County, Huzhou City, Zhejiang Province, and Shandong Yuwang Ecological Foodstuffs Co., Ltd., respectively. Potassium hydroxide (KOH) and urea ($\text{CO}(\text{NH}_2)_2$) were purchased from Sinopharm Chemical Reagent Co. *N,N'*-Methylenebisacrylamide (MBA) and acrylic acid (> 99% (GC), containing 180–200 ppm MEHQ stabilizer) amine were purchased from Shanghai McLean Biochemistry Science and Technology Co., Ltd. Potassium persulfate (KPS) was purchased from Daimao Chemical Reagent Co., Ltd. Soil samples were collected from the third living area of Henan Agricultural University. The soil depth was 0–15 cm (plow layer), and the soil was air-dried and then sieved through a 2-mm sieve and stored.

2.2 Preparation of slow-release fertilizers

2.2.1 Preparation of bamboo biochar

The bamboo material was soaked in distilled water for 24 h to remove impurities and dirt and then dried in an oven at 105 °C. The dried bamboo was placed in a tube furnace, heated to 700 °C at a heating rate of 10 °C/min, and charred under a constant N_2 flow rate of 30 ml/min for 2 h. The carbonized bamboo was ground to 80 mesh and stored in a sealed bag for use.

2.2.2 Preparation of bamboo biochar-hydrogel composites

The reaction of soy protein (SPI) with acrylic acid (AA), potassium persulfate (KPS), and *N,N'*-methylenebisacrylamide (MBA) was used to synthesize soy protein-based hydrogels through the graft polymerization method. Firstly, 5 g of soy protein (SPI) was completely dissolved with 0.45 g of KOH in a beaker containing 35 ml of deionized water, which was placed on a magnetic stirrer and reacted at 70 °C for 1.5 h to obtain the soy protein alkaline solution. Next, 13.4 g of KOH was dissolved in 31 ml of deionized water. Once the solution was cooled to room temperature, 15 g of acrylic acid (AA) was added to prepare a potassium acrylate solution. Subsequently, the soy protein alkaline solution was mixed with the potassium acrylate solution, and urea was then added in amounts of 0, 10, and 15 g, corresponding to ratios of 1:0, 1:2, and 1:3 with SPI, respectively. These hydrogel precursor solutions were named SPB-0, SPB-2, and SPB-3. The hydrogel precursor solutions loaded with different urea concentrations were then supplemented with 0 g and 0.2 g of bamboo biochar, representing 0% and 4% of SPI, respectively. These six resulting mixtures were sonicated using an ultrasonic cell crusher with 450 W power for 10 min to make a homogeneous mixture and labeled as SPB-0/2/3-(0%/4%). Finally, 0.15 g of MBA and 0.275 g of KPS were dissolved in 14 ml of distilled water and added to the six mixtures described above and placed on a magnetic stirrer at 70 °C and stirred until gelation, polymerization, and cross-linking occurred for 30 min. After allowing it to cool to room temperature, the hydrogel was removed from the beaker, cut into the desired sizes, and washed several times with anhydrous ethanol to remove unreacted monomers and low-molecular-weight polymers. The hydrogel was then dried in an oven at 45 °C for 2 days and stored in a desiccator until use. The sample names and the corresponding urea and biochar loadings are shown in Table 1.

2.3 Characterization of SPB

The FTIR spectra of soy protein (SPI), bamboo biochar, and the six SPB samples were determined in the range of

Table 1 Raw material composition of the SPB materials

Samples	SPI (g)	AA (g)	MBA (g)	Urea (g)	Bamboo biochar (g)
SPB-0–0%	5	15	0.15	-	-
SPB-0–4%	5	15	0.15	-	0.2
SPB-2–0%	5	15	0.15	10	-
SPB-2–4%	5	15	0.15	10	0.2
SPB-3–0%	5	15	0.15	15	-
SPB-3–4%	5	15	0.15	15	0.2

4000–400 cm^{-1} using an infrared spectrometer (Nicolet Avatar 360, USA). The SPBs were analyzed by X-ray diffraction (XRD) using an XRD diffractometer (Rigaku Ultima IV, Japan) to determine their chemical structure. The C1s, N1s, and O1s peak spectra on the surface of the six SPB samples were observed by X-ray photoelectron spectroscopy (Kratos Axis Ultra DLD, UK). The surface morphology and internal pore structure of SPB and biochar were examined using an SEM (Hitachi, Japan). The thermal stability of SPB materials was determined by TGA (NETZSCH TG 209 F3 Tarsus, Germany) at a ramp rate of 10 $^{\circ}\text{C}/\text{min}$ with a constant N_2 flow and a temperature range of 25–800 $^{\circ}\text{C}$.

2.4 Swelling capacity of SPB

Under room temperature conditions, 0.2 g of six dried SPB samples was loaded into tea bags (100 mesh) and weighed as M_i , which were subsequently immersed in deionized water solution at pH 5, 7, and 9 to investigate their swelling properties under different pH conditions. Once the SPB adsorption is saturated, the tea bags were removed and weighed after being allowed to stand for 10 min to remove gravitational water. The weight of the samples was recorded as M_f . The water absorption rate (SWR) was calculated using the following equation [33]:

$$\text{SWR} = \frac{M_f - M_i}{M_i}$$

2.5 Water-holding capacity of SPB

The bottoms of the plastic flower pots were lined with filter paper, and each pot was filled with 200 g of soil. Then, 2% of each of the six SPBs was added to different flower pots; one pot with no additional material was used as a control. After thoroughly mixing the SPBs with the soil, their weights were recorded as W_0 . Subsequently, water was slowly added to the pots until it began dripping from the bottom, and the pots were re-weighed and recorded as W_1 . The water-holding capacity (WH%) of SPB was calculated as follows [34]:

$$\text{WH} = \frac{W_1 - W_0}{W_0} \times 100\%$$

2.6 Evaluation of SPB water retention

The SPB samples were thoroughly mixed with 100 g of dry soil (particle size less than 2 mm) in a beaker at 2% addition and weighed as M_0 . An additional beaker containing only soil was used as a control, with three replicates for each sample. Subsequently, 100 ml of deionized water was added to the beaker and weighed as M_1 . The beaker was then stored at room temperature, and the weights (M_i) were recorded on days 0.5, 1, 2, 3, 4, 7, 13, 19, 24, and 30. The water retention of SPB (WR%) was calculated as follows [35]:

$$\text{WR} = \frac{M_i - M_0}{M_i - M_0} \times 100\%$$

2.7 Slow release of nitrogen fertilizers

In this study, the slow release of urea in SPB was tested using the water dissolution rate method in the following steps: First, 0.5 g of dried SPB material was added to a beaker containing 300 ml of deionized water, serving as a medium for N fertilizer release. The beaker was sealed with plastic wrap to prevent evaporation of the solution. Ten milliliters of aqueous solution was taken from the beaker on days 0.5, 1, 3, 4, 7, 13, 19, 24, and 30 used for the determination of urea content. Before sampling, the solution was stirred with a glass rod for 30 s, and 10 ml of deionized water was subsequently added to the beaker to maintain the solution volume constant. The urea content in the aqueous solution was determined using the p-dimethylaminobenzaldehyde method [36], where the absorbance was measured at 440 nm using a standard urea curve. The slow-release rate of urea was calculated as follows [37]:

$$\text{Cumulative release} = \frac{M_t}{M_i} \times 100\%$$

where M_t is the amount of urea released at different moments in the aqueous solution and M_i is the amount of urea loaded in the SPB.

2.8 Pot experiment

2.8.1 SPB-0–0% plant culture and drought mitigation

Two sets of replicated experiments were conducted using SPB-0–0% at a 2% addition rate to the potting soil, with each set comprising three replicates. First, 320 g of soil was thoroughly mixed with 6.4 g of SPB-0–0% material in

plastic pots. For the control group, no material was added to the pots. Subsequently, eight cucumber seeds (*Cucumis sativus* L. cv. Jinyou No. 4) of similar size were sown in each pot, and 150 ml of deionized water was added to each pot, which was placed in a light incubator set at 25 °C and 60% humidity for growth. Thirty milliliters of deionized water was added to the pots every 3 days. At 20 days of growth, the cucumber seedlings in one group of potting experiments were taken out and rinsed with deionized water. The germination rate, root length, plant height, fresh weight, dry weight, and chlorophyll content of the cucumbers were measured. In the other group of potting experiments, watering was ceased, and the morphology of the cucumber seedlings was observed, photographed, and recorded after 5 days.

2.8.2 Plant culture of different SPB materials

The pot experiment was carried out by adding each of the six SPBs (SPB-0–0/4%, SPB-2–0/4%, and SPB-3–0/4%) at 0.5% (1.6 g) to 320 g of soil, while soil without material addition called CK, soil with the same urea additions as SPB-2–0/4% and SPB-3–0/4% loadings called Urea-2 and Urea-3, were used as controls. To avoid the influence of seed quality problems, only germinated seeds were used in subsequent experiments. Three germinated seeds were added to each pot, followed by 150 ml of deionized water. The incubation conditions, growth duration, and measurement indexes were the same as those in subsection 2.6.1.

2.8.3 Plant culture with different SPB concentrations in soil

In order to study the effects of different SPB concentrations on plant growth, SPB-2/3–4% was mixed with 320 g of soil at concentrations of 0.25%, 0.5%, and 0.75%. Each pot was then planted with three germinated cucumber seeds and irrigated with 150 ml of deionized water. The pots were placed in the light incubator under the same incubation conditions, growth duration, and measurement indexes as described in subsection 2.6.1.

3 Results and discussion

3.1 Preparation and structural characterization of soy protein-based hydrogel (SPB) materials

3.1.1 Preparation mechanism of SPB materials

Under alkaline conditions, soy protein undergoes hydrolysis, exposing reactive groups, such as -OH, -COOH, and -NH₂. Meanwhile, the initiator, potassium persulfate (KPS), decomposes at high temperatures, producing a large number of sulfate ions (-SO₄²⁻), which can interact with H atoms

from the -OH of the soy protein to produce active groups. Acrylic monomers can undergo grafting reactions with the reactive groups of soy protein, while chain polymerization reactions occur among their monomers, promoting the growth of molecular chains to obtain high polymer polymers (SPI-PAA). At the same time, these active groups can react with MBA, serving as a cross-linking agent, to form a three-dimensional cross-linked network. Throughout this process, urea and biochar act as nutrients and dispersants, participating in the entire polymerization reaction without interfering with the free radical polymerization process. This enriches the SPB network structure, enhances the water absorption of SPB, and prolongs the nutrient release time. The soy protein-based hydrogel (SPB) material preparation mechanism is shown in Fig. 1.

3.1.2 FTIR, XRD, and TG analyses of SPBs

The functional groups of bamboo charcoal, soy protein (SPI), and six soy protein-based hydrogels (SPB) were investigated using Fourier infrared transform spectroscopy. The absorption peak at 3332 cm⁻¹ in Fig. 2a is attributed to the stretching vibration of the amide I band in soy protein (SPI), whereas that of 1690–1750 cm⁻¹ corresponds to the stretching vibration of C=O [38], which can be attributed to the stretching vibration of the amide II band. The absorption peak at 1550 cm⁻¹ corresponds to the stretching vibration of the amide III band, which can be attributed to the deformation vibration of N-H in the amide bond of SPI. These vibration peaks prove the formation of protein-based hydrogel materials [39]. In Fig. 2b, the absorption peaks at 2860–2980 cm⁻¹ are attributed to the C-H vibration due to the vibration of the vinyl group (-CH-CH₂) in acrylic acid. The absorption peaks observed at 769–920 cm⁻¹ result from the out-of-plane bending vibration of O-H, which is due to the reaction of the vinyl group in acrylic acid with water to form alcohols during the hydrogel graft copolymerization process. It was observed that the loading of bamboo charcoal in the SPB material enhanced the tensile vibration of the C-H bond at the same urea loading, while the C=O vibration from the soy protein was weakened. This change can be attributed to bamboo charcoal introducing the reactive group -OH in the SPB material, which provided more active sites for cross-linking reactions and promoted the cross-linking between acrylic acid and soy protein. Meanwhile, the increase of hydrophilic groups as C-H is favorable for SPB water retention properties. The absorption peaks at 1400–1510 cm⁻¹ are due to the stretching vibration of COO⁻ [40, 41], suggesting that the reactive groups (carboxyl groups) in the soy protein molecules are esterified with the by-products (alcohols) of acrylic acid during the gelation process. These reactions lead to cross-linking, confirming the successful grafting of acrylic acid onto the soy protein

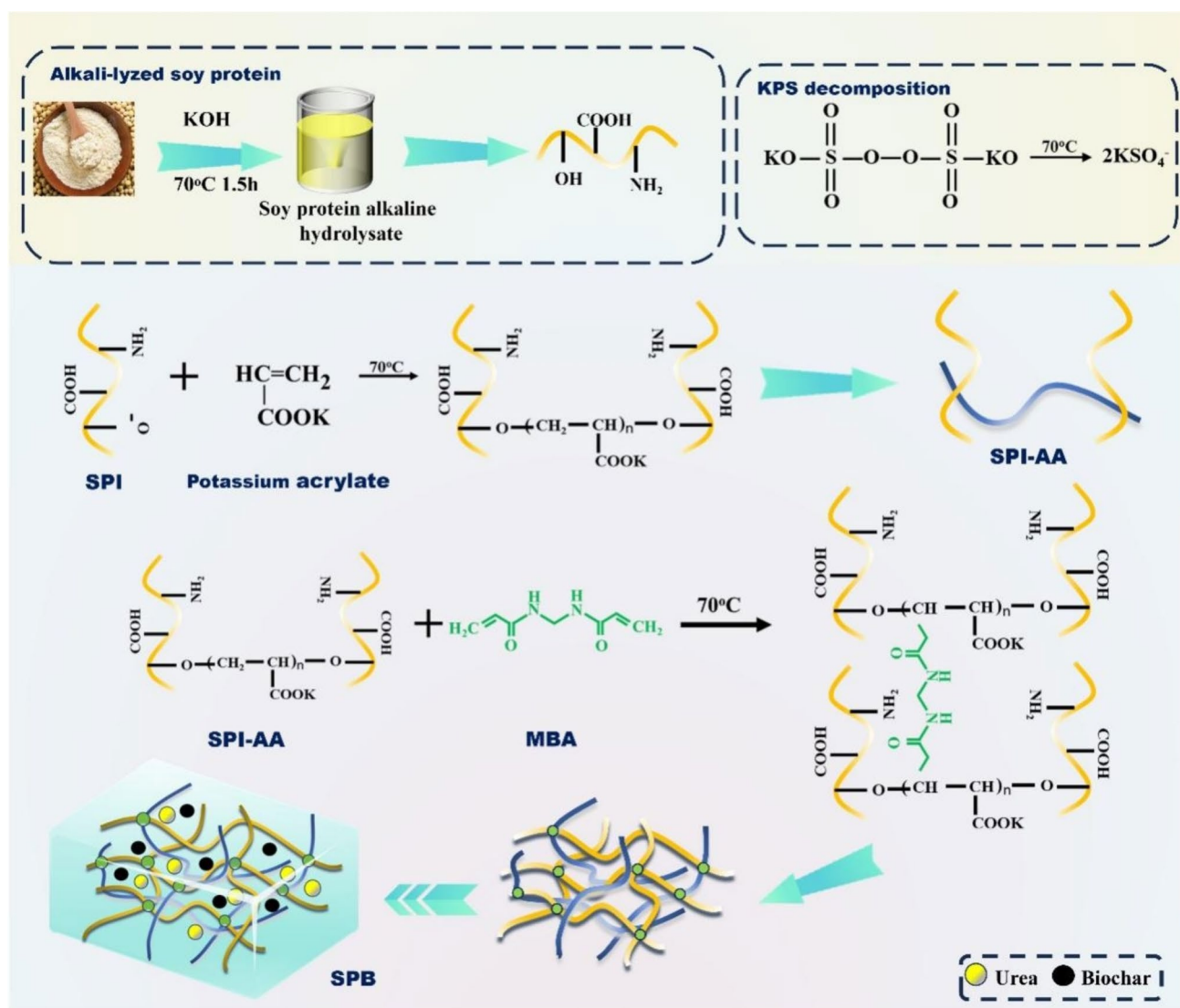


Fig. 1 SPB preparation mechanism

molecules [42, 43]. The absorption peaks at 3420 cm^{-1} and $1050\text{--}1250\text{ cm}^{-1}$ in Fig. 2b correspond to the stretching vibration of -NH_2 [33] and C-N, respectively. These vibrations were enhanced with the increase in urea loading, which indicates the successful loading of urea into the hydrogel material.

The XRD spectrum of the SPB material is shown in Fig. 2c. The crystal diffraction peaks of polyethylene (glutamic acid) methanol are detected at $2\theta = 27.34^\circ$ and 33.71° , which further proves the successful grafting of acrylic acid and soybean protein. This observation also aligns with the cross-linking mechanism indicated in the FTIR spectra. In addition, the characteristic diffraction peaks of urea ($\text{CO}(\text{NH}_2)_2$) were detected in the urea-loaded SPB materials (SPB-2-0%, SPB-2-4%, SPB-3-0%, and SPB-3-4%). The intensity of the peaks increased with the increase in urea

loading, which indicated that urea was successfully loaded in the SPB materials.

Figure 2d shows the results of the thermogravimetric analysis of six SPB materials, from which it can be seen that the thermal decomposition of SPB materials can be divided into three stages. The first stage of thermal decomposition occurs between 35 and 150°C and is attributed to the release of free water from the composite [44]. The second stage, between 150 and 490°C , results in mass loss due to the generation of ammonia, CO , and CO_2 from the unstable functional groups in the acrylic acid and soy protein molecules [42, 43]. The third stage of thermal decomposition takes place between 490 and 760°C , which is due to the breakage of the main chain of soy protein and the decomposition of acrylic acid [35, 45]. Meanwhile, the thermogravimetric analysis results indicate that when the temperature exceeds

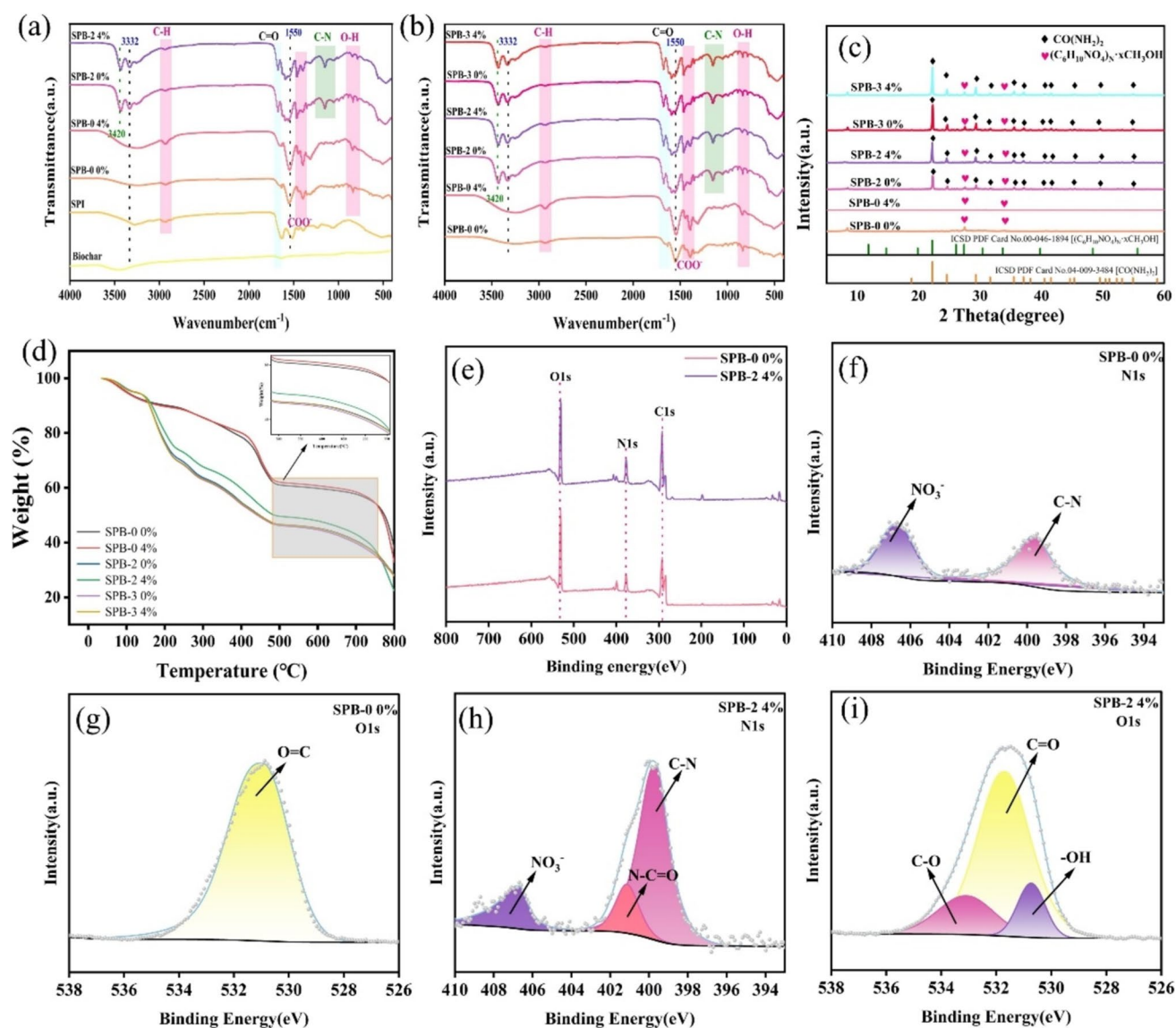


Fig. 2 A, b FTIR spectra of biochar, SPI, SPB-0-0%, SPB-0-4%, SPB-2-0%, SPB-2-4%, SPB-3-0%, and SPB-3-4%; c XRD spectra of six SPBs; d thermogravimetric curves of six SPBs; e XPS spectra

of SPB-0-0% and SPB-2-4%; f, g peaks of N1s and O1s for SPB-0-0%; h, i peaks of N1s and O1s for SPB-2-4%

150 °C, the decomposition rate of urea-loaded SPB materials (SPB-2-0%, SPB-4%, SPB-3-0%, SPB-3-4%) was significantly higher than that of the unloaded SPB materials (SPB-0-0%, SPB-0-4%). Moreover, the decomposition rate increased with the increase in urea loading, which is due to the poor thermal stability of urea, decomposing at high temperatures to form biuret, triuret, cyanuric acid, and ammonia. This observation aligns with the results of Huang and Priya et al. [37, 46]. In addition, at the same urea loading, the decomposition rate of biochar-added SPB materials was lower than that of non-biochar-added SPB materials, suggesting that adding biochar helps improve the thermal stability of the materials [47].

3.1.3 XPS analysis of SPBs

The chemical states and compositions of SPB materials were investigated using XPS spectroscopy. The peaks at 291.10 eV, 377.38 eV, and 527.31 eV in Fig. 2e represent the C1s, N1s, and O1s spectra, respectively. As shown in Fig. S1a (Supporting Information), peaks at 284.80 eV, 288.09 eV, 292.54 eV, 285.98 eV, and 295.37 eV were detected in the XPS spectra of the C1s of all the six SPB materials, which corresponds to the OH-C=O, C=O, C-O, C-N, and C-C bonds, respectively. The OH-C=O and C=O bonds were derived from the acrylic monomer in the SPB material [41]. The C-N bond originated from the amino

group in soy protein [48], while the C-O and C-C bonds were formed during the graft copolymerization of soy protein and acrylic acid. In Fig. 2f, peaks at 399.54 eV and 406.82 eV corresponding to NO_3^- and C-N were observed, respectively. Additionally, Fig. 2h reveals the formation of a new N-C=O bond in the urea-loaded material, detected at 400.36 eV [47]. The observation from Fig. S1b (Supporting Information) shows that the molar content of C-N and N-C=O bonds increases with the increase in urea loading, which is due to the fact that urea mainly binds to the backbone of soy protein in the form of C-N bonds. With the increase in urea loading, urea also binds to the cross-linking sites via N-C=O bonds. The peak at 531.32 eV in Fig. 2g can be attributed to the O=C bond. Additionally, from Figs. 2h and S1 (support information), the peaks at 530.68 eV and 532.55 eV associated with the C-O and -OH bonds, respectively, appeared after the introduction of urea. This may be caused by the aggregation of urea on the surface of the material, which reduces its active reaction sites, and

the self-polymerization of some low-molecular-weight substances [49]. These findings further demonstrate the successful grafting of soy protein with acrylic acid and the effective loading of urea.

3.1.4 SEM analysis of SPBs

Figure 3b shows the SEM image of biochar, which exhibits a typical porous structure favorable for the absorption and storage of nutrients and water. From Fig. 3c, the SPB material unloaded with urea and biochar exhibited an obvious 3D porous network structure, which further proves the successful grafting of soy protein with acrylic acid. Comparing Fig. 3c, d, it can be seen that the loading of biochar provides more active sites for the reaction, which further enriches the network structure of the polymer and makes its structure denser. Additionally, Fig. 3d, f, and h shows that the biochar was uniformly distributed in the interior and the surface of the material,

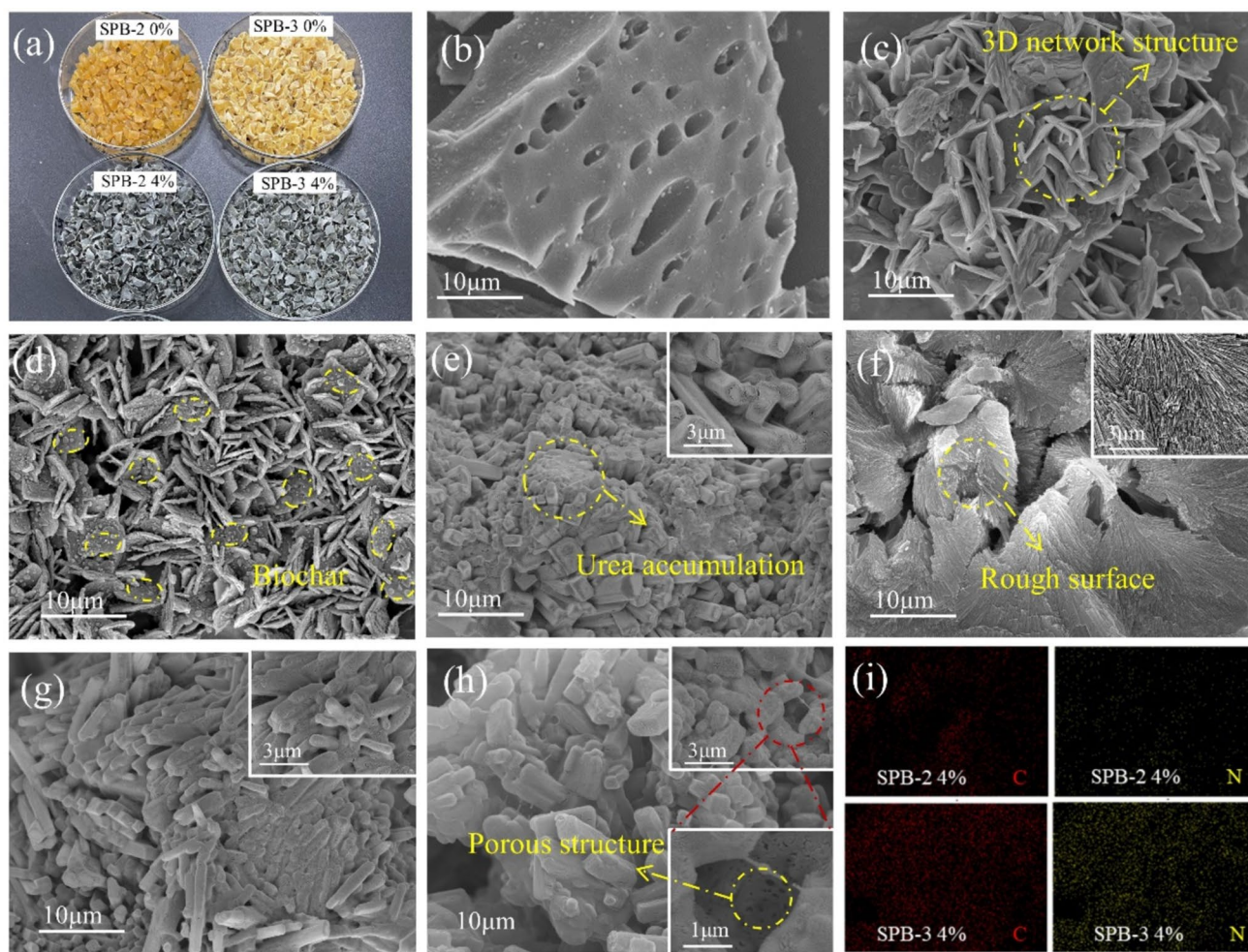


Fig. 3 a Photographs of four SPB materials; SEM images of b bamboo charcoal, c SPB-0-0%, d SPB-0-4%, e SPB-2-0%, f SPB-2-4%, g SPB-3-0%, and h SPB-3-4%; and i elemental mapping images of C and N in SPB-2-4% and SPB-3-4%

which makes the surface rougher and further increases the specific surface area of the material. This increased roughness and surface area promote the slow release of fertilizers and water more effectively [50]. In addition, by comparing Fig. 3c, e, it is evident that urea loading resulted in an increase in the roughness of the material. This phenomenon can be attributed to the precipitation and aggregation of urea on the surface of the material, as well as its attachment to the interior of the material due to strong hydrogen bonding with soybean proteins. Meanwhile, a comparison of Fig. 3e, g reveals that as the amount of urea loading increases, the lamellae structure in the SPB material becomes denser, and the attachment of urea on its surface becomes uniform. From Fig. 3h, it can be seen that SPB-3–4% has formed a complex hierarchical structure, and its local magnification to 1 μm can be observed that there are different sizes and dense pore-like structures between the layers. This phenomenon proves the formation of a three-dimensional porous network structure in SPB and suggests that it has great potential in water retention and slow release. The elemental mapping results of SPB-2–4% and SPB-3–4% are shown in Fig. 3i, which further demonstrates the uniform distribution of biochar and urea in the SPB materials.

3.2 Swelling capacity of SPB at different pH as well as water-holding and water-retaining properties in soil

As hydrogel has a high water absorption and retention capacity, its internal cross-linking network can be unfolded to facilitate the absorption of more water, as shown in Fig. 4a. During the process of water absorption, the hydrogel can retain a large amount of water as well as do not dissolve themselves. Hence, it can be used as a soil conditioner to enhance the water content of the soil. Therefore, in arid and semi-arid areas with insufficient rainfall, hydrogel has a great potential for application [51]. Figure 4b shows the expansion rate of SPB material at different pH. The figure reveals that regardless of pH levels, the expansion rate of SPB material can reach more than 222.3 g/g. The variation in the expansion rate at various pH was relatively small, which indicates that the SPB material has a wide range of applicability in both saline and acidic soil conditions. In addition, the swelling rates of SPB materials with bamboo biochar (SPB-0–4%, SPB-2–4%, and SPB-3–4%) were higher than those without biochar (SPB-0–0%, SPB-2–0%, and SPB-3–0%) under the same conditions. This indicates that the introduction of biochar could enhance the swelling rate of the materials, which was attributed to the porous structure of biochar and its abundant hydrophilic functional

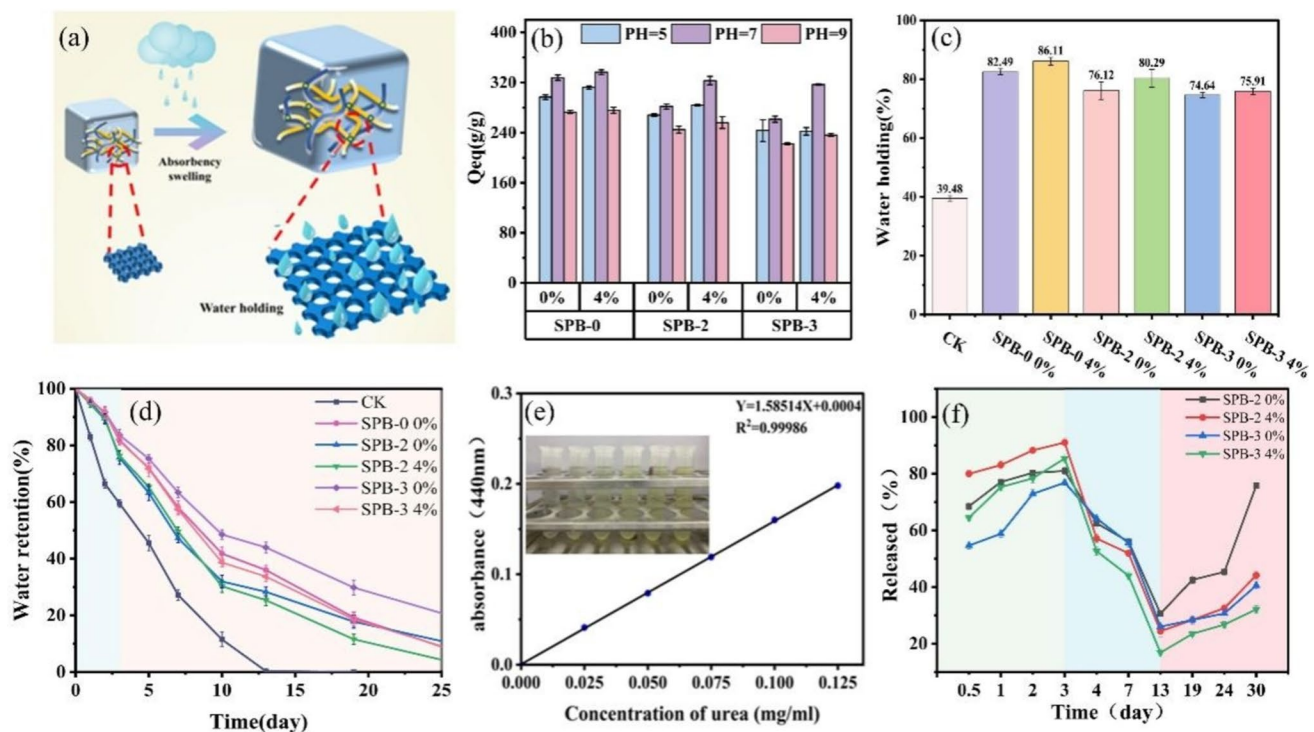


Fig. 4 a Schematic diagram of SPB water absorption; b SPB swelling rate at pH=5, 7, and 9; c water-holding capacity of soil when SPB was added at 2% (CK is soil without SPB material); d water retention

capacity of SPB; e standard curve of urea; and f N fertilizer release rate of SPB material loaded with urea

groups (C-H, O-H groups) [50, 52], which facilitate the absorption of water molecules due to surface tension and interaction with the surface functional groups. The SEM images showed that the SPB material containing bamboo biochar exhibited a denser network structure, which can be seen from Fig. 2b that the addition of biochar enhanced the stretching vibration peaks of the hydrophilic groups C-H in the SPB. In contrast, the stretching vibration of the C=O bond and the -OH group in the biochar was weakened, which suggests that the introduction of the biochar produced more cross-linking sites. At the same time, the SPB expansion rate decreases with the increase in urea loading, which is due to urea binding within the main chain of soybean protein and replacing hydrophilic groups in the soybean protein. In addition, increasing urea loading is equivalent to reducing the concentration of acrylic acid, which affects the subsequent polymerization and cross-linking of soybean protein and acrylic acid, weakening the expansion rate of SPB [53]. The observed highest swelling rate of the SPB material at pH 7 can be attributed to the protonation reactions of -COO- groups in an acidic environment, which enhances the hydrogen bonding between carboxyl groups and inhibits electrostatic repulsion between anions, thus affecting its swelling property [54]. In contrast, under alkaline conditions, the presence of cations in the alkaline solution can shield the -COO- groups and inhibit the electrostatic repulsion between the anions, which weakens the expansion of the polymer network [55], allowing the polymer network to fully extend and reach its maximum swelling rate (higher than 261.5 g/g) under neutral conditions.

As shown in Fig. 4c, the water-holding properties of SPB materials in soil showed the same trend, indicating that the introduction of biochar uniquely enhances the water-holding capacity of SPB. The water-holding rate of soil with SPB materials was significantly increased, exceeding 29% compared to the control. This enhancement is due to the hydrophilic nature of the SPB material, which colloiddally swells upon contact with water. Additionally, the internal cross-linking network unfolds, allowing the material to retain a large amount of water within its porous network. Soil treated with SPB-0-0%, SPB-0-4%, and SPB-2-4% materials exhibited increased water-holding capacities by 108%, 118%, and 103%, respectively, compared to the control. Moreover, regardless of the urea loading, the water-holding capacity of soil applied with biochar-added SPB materials (SPB-0-4%, SPB-2-4%, SPB-3-4%) was slightly higher than those treated with SPB materials without biochar (SPB-0-0%, SPB-2-0%, SPB-3-0%). This difference is attributed to the enrichment of the cross-linking network and the provision of more hydrophilic functional groups due to the introduction of biochar. The water-holding capacity of the soil with SPB material decreased with the increase in urea loading, with SPB-3-4% showing a decrease of 10.2%

compared to SPB-0-0%. This reduction can be attributed to the high concentration of urea affecting the cross-linking within SPB. In summary, the SPB material provided the soil with higher water-holding capacity due to its excellent swelling properties, and the addition of biochar further enhanced its performance.

Figure 4d shows the water retention of the soil. It was observed that the soil without SPB lost all its water on the 13th day, whereas the water retention of the soil with SPB material remained above 25.36%. The water-holding capacity of the soil treated with SPB-3-0% was 43.9%. It is shown that the use of SPB can significantly reduce the evaporation rate of soil moisture. This effect stems from the viscous and ductile nature of the SPB material, which enhances the adhesion of soil aggregates, thereby reducing the surface area of the soil and improving its water retention [35]. Initially, the water retention rate of soil treated with SPB material showed similar changes in the first 3 days. However, after this period, the water retention of soil treated with SPB-0-0%, SPB-2-0%, and SPB-3-0% was higher than that of SPB-0-4%, SPB-2-4%, and SPB-3-4%, which was attributed to the introduction of biochar in the SPB material, accelerating its degradation rate. The porous structure, ash content, and minerals present in bamboo biochar provide more habitats and nutrients for microbial growth [56]. These microorganisms promote the degradation of the SPB materials, destroying its internal structure and reducing its weight, thus decreasing the water retention capacity of the soil. However, the degradation of soybean protein within the SPB material can serve as a natural nitrogen fertilizer, enriching the soil with nutrients that plants can absorb and utilize to promote plant growth. Therefore, in SPB materials with little difference in water retention, the addition of bamboo biochar is more beneficial to plant growth. In addition, it was found that the water retention of the soil increased with the increase of urea loading. This can be attributed to the introduction of amide groups through urea loading, which increases the hydrophilic groups, thereby enhancing the water retention capacity of SPBs [2, 57]. For instance, on the 25th day, the water content of the SPB-applied soil could still be maintained between 8 and 20%, which indicates that the SPB material has great application in soil water conservation and management.

3.3 Slow-release behavior of nitrogen fertilizers in various SPBs

Figure 4e shows the standard curve for the determination of urea content in solution as determined by the p-dimethylaminobenzaldehyde method [58]. Figure 4f depicts the rate of urea release from various SPB materials over a period of 30 days. As shown in the figure, urea release from SPB can be divided into three stages: an initial rapid release in

the first 3 days, followed by a gradual decrease in urea concentration from days 3 to 13, and a slower release after day 13, with the urea release from SPB materials less than 46% at day 24. On day 30, the SPB-2-0% material exhibited the highest urea release rate, reaching 75%. Compared to pure urea that dissolves completely in water for only 1 day [59], the SPB material significantly reduced the rate of nitrogen fertilizer release. The rapid release of urea in the first stage can be attributed to water penetrating the interior of the polymer, which results in the gradual release of urea from the material into the solution. For the same urea loading, the rate of urea release from the SPB materials with the addition of bamboo biochar (SPB-2-4%, SPB-3-4%) was higher than that of the materials without bamboo biochar (SPB-2-0%, SPB-3-0%). This is consistent with Wu et al.'s study suggesting that biochar promotes nutrient release during the expansion of the SPB material, which gradually releases the urea-bound biochar inside [50]. The decrease in urea concentration in the second stage is due to the adsorption of urea by the porous structures in the biochar and hydrogel. Adsorption equilibrium was reached on the 13th day, with the SPB material containing bamboo biochar demonstrating a better adsorption effect, which is attributed to the denser porous structure inside the material. In the third stage, the soy protein-acrylic acid network in SPB may undergo gradual hydrolysis to oligomers, which promotes the slow release of urea. Furthermore, the figure indicates that the SPB materials with added biochar (SPB-2-4%, SPB-3-4%) exhibited an excellent urea slow-release effect. This enhancement can be attributed to the addition of biochar, which provides more reactive sites for polymerization reactions, significantly increasing the specific surface area of the material. Also, the oxygen-containing functionalities, as well as hydrogen bonding and electrostatic interactions present in the active groups of biochar, can slow down the exchange release rate of N in water. The synergistic effect of hydrogel and biochar further enhances the slow-release effect of SPB materials [60]. In summary, SPB materials can slow the rate of fertilizer release, presenting great research significance and practical application in improving fertilizer utilization and reducing N fertilizer loss.

3.4 Effect of SPB materials on cucumber growth and mechanism of action

3.4.1 Effect of SPB-0-0% on germination and growth of cucumber

In order to assess the biotoxicity and growth effectiveness of the SPB material, pot experiments were conducted using SPB-0-0%. From Fig. 5a-c, it can be seen that the application of SPB-0-0% was able to promote plant growth and enhance the germination rate of cucumber seeds, resulting

in a 25% increase. Moreover, as shown in Fig. 5g-i, the plant height, root length, fresh weight, dry weight, leaf area, and relative chlorophyll content of SPB-0-0%-treated cucumber seedlings significantly increased by 53.72%, 32.77%, 121.2%, 66.67%, 88.67%, and 49.63%, respectively, compared to the control. From Fig. 5d, it was evident that the root system of the cucumber could pass through the hydrogel and grow normally within it, proving that the SPB material had good biocompatibility and could promote root growth. To verify the water retention effect of SPB materials, watering was stopped after the cucumber seedlings had grown for 20 days. Subsequently, the cucumbers untreated with SPB-0-0% showed obvious wilting and collapse after 5 days without watering, while the SPB-treated cucumbers continued to grow, as shown in Fig. 5f. This is because the porous structure of SPB-0-0% enables it to retain water more effectively, while the expansion of the hydrogel after water absorption can increase the porosity of the soil [61]. At the same time, the small molecules produced by the degradation of soybean proteins can be used as a natural nitrogen fertilizer to promote the growth of the plant and nutrient absorption [62]. Therefore, SPB materials have great application value in mitigating drought stress and promoting plant growth.

3.4.2 Effects of different SPB and different additive amounts on the growth of cucumber

In order to further verify the effects of the introduction of bamboo biochar on the slow-release effect of SPB materials and plant growth, pot experiments were conducted with various SPB materials. Figure 6a illustrates that the addition of urea (Urea-2/3) in the soil can promote the growth of cucumber. This effect is due to the proper use of nitrogen fertilizer, which can enhance the activity of photosynthesis-related enzymes in plants and improve the photosynthetic capacity of the leaves, thereby promoting overall plant growth [63]. From Fig. 6b-e, it can be seen that the cucumber growth in the soil treated with SPB material was better than the untreated soil. The fresh weight of cucumber in the soil treated with SPB material increased by 57.69~139.32% compared with the control due to the high water retention of the hydrogel material and the excellent slow-release effect. In addition, cucumbers grown in SPB material with added biochar exhibited superior growth than those grown in SPB material without added biochar at the same urea loading. After SPB-2-4% soil treatment, the cucumber root length, plant height, fresh weight, dry weight, relative chlorophyll content, and leaf area increased by 42.45%, 19.96%, 50%, 32.59%, 28.83%, and 16.28%, respectively, as compared to SPB-2-0% treatments (with the same urea loading). With increased urea loading, SPB-3-4% showed a similar trend to SPB-3-0% treated cucumber. This can be attributed to the addition of bamboo charcoal, which further enriched

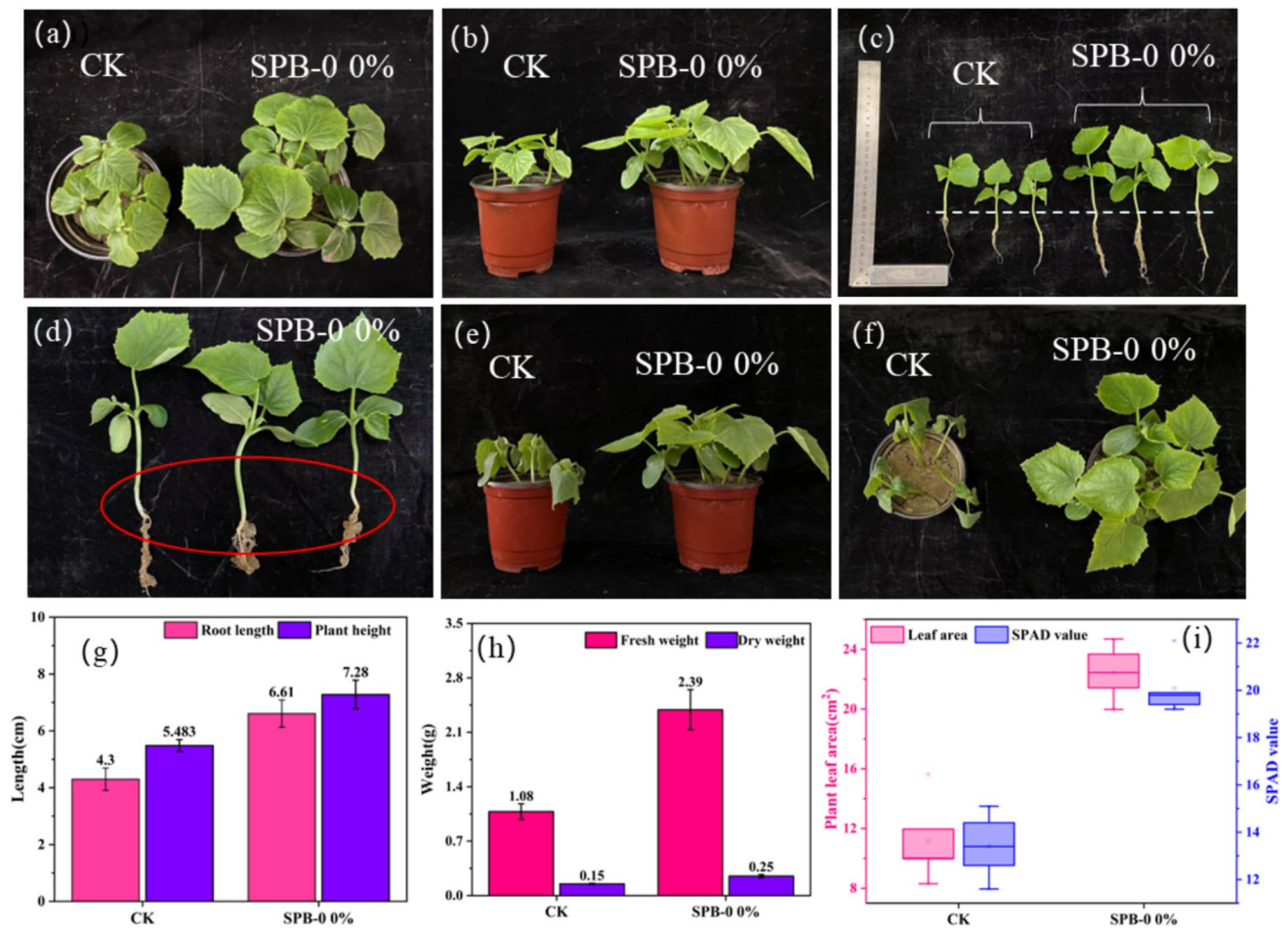


Fig. 5 a–d Growth of cucumber in soil treated with SPB-0-0% and untreated soil (CK), e, f morphology of cucumber in soil treated with SPB-0-0% and untreated soil (CK) 5 days after watering cessation;

in soil treated and untreated (CK) with SPB-0-0%: g root length and plant height, h dry and fresh weights, i leaf area and relative chlorophyll content of cucumber

the network structure of SPB and optimized its performance. Additionally, biochar could provide microorganisms with habitat and nutrients to enrich the microbial community structure [56], which could increase the biological abundance of the nitrogen cycle and enhance the nitrogen content of the soil [64]. Thus, it aligns the supply of water and fertilizer more closely with plant growth patterns and provides more benefits for plant growth. Meanwhile, morphological indicators of cucumber showed that elevated urea loading would somewhat inhibit plant growth [65]. As shown in Fig. 6b, c, the root length, plant height, and fresh dry weight of Urea-3-treated cucumber were lower than those of Urea-2-treated cucumber. However, the plant height of cucumber treated with SPB-3-0% and SPB-3-4% at the same urea loading increased by 14.38% and 15.75%, respectively. This further proved that the SPB material has an excellent slow-release effect, thus avoiding the inhibition of plant growth caused by the high nitrogen content in the soil. In summary, the SPB material with bamboo biochar

can further promote the growth of cucumber, which can be attributed to the enrichment of the network structure of the hydrogel material and the slow release of nutrients caused by the addition of the bamboo biochar. Moreover, the nutrients in biochar can promote the growth of beneficial microorganisms and plants [6].

From the above experiments, it was evident that SPB materials can promote the growth of cucumber. In order to ascertain the effect of varying additive amounts on the growth of plants, this study examined the best slow-release effect of SPB-2-4% and SPB-3-4% when applied to the soil at additive amounts of 0.25%, 0.5%, and 0.75%. From Fig. S2 (Supporting Information), the growth of cucumber was enhanced with increasing SPB addition. The increase in root length, plant height, fresh dry weight, relative chlorophyll content, and leaf area of cucumber exhibited a significant and positive correlation with increasing SPB addition (specifically SPB-2-4%, Fig. 6f–i). For instance, compared to soil treated with 0.25% addition, the soil with 0.75% SPB

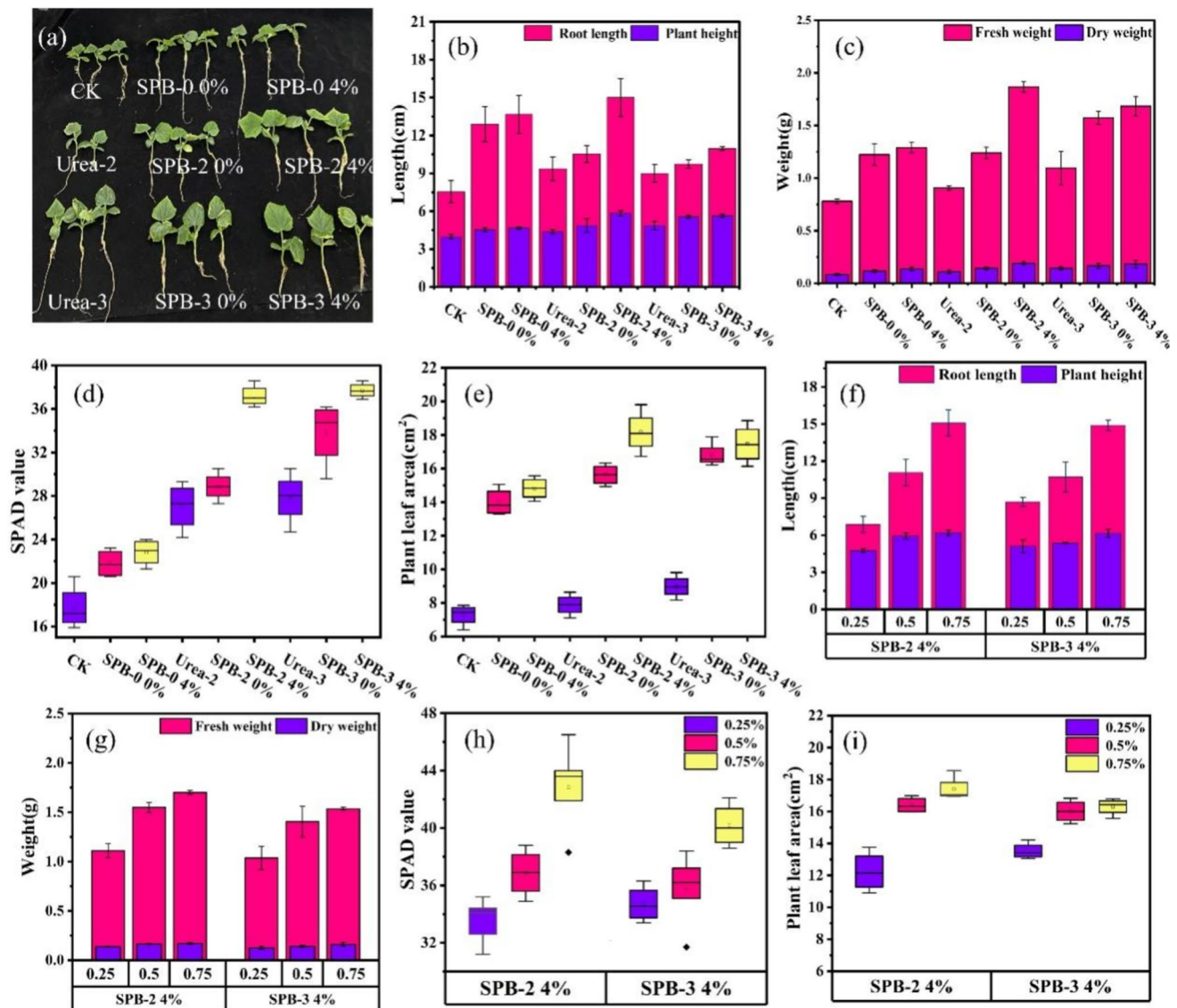


Fig. 6 a Growth of cucumber in soil (5% addition) spiked with SPB-0/2/3-0%, SPB-0/2/3-4%, Urea-2/3 (consistent with SPB-2/3 urea content) and untreated soil (ck). After treatments, root length and plant height (b), dry and fresh weights (c), relative chlorophyll con-

tent (d), leaf area (e); with different additions of SPB-2/3-4%, the cucumbers' root length and plant height (f), dry and fresh weights (g), relative chlorophyll content (h), leaf area (i)

addition increased the cucumber root length, fresh weight, leaf area, and relative chlorophyll content by 119.60%, 53.15%, 42.16%, and 31.34%, respectively. Thus, bamboo biochar-hydrogel composites have great potential for agro-nomic and horticultural applications.

3.4.3 Mechanism of action of SPB materials in promoting the growth of cucumber seedlings

Nitrogen is the most crucial nutrient element for plant growth and development. It not only serves as a substrate for synthesizing chlorophyll, nucleic acids, and proteins in plants but also aids in fostering various physiological and

metabolic processes, thereby facilitating plant growth and development [66, 67]. Li's study showed that excessive N fertilization caused chloroplast enlargement and inhibition of RuBisCo enzyme activity, leading to a decrease in photosynthetic nitrogen use efficiency (PNUE) [68]. At the same time, moisture serves as a fundamental prerequisite for plant growth and development, whereas drought causes elevated levels of reactive oxygen species (ROS) in plants. When ROS content is within the plant's tolerance range, antioxidant metabolism can be regulated to avoid damage to the plant. However, if the level of ROS is too high, it can damage the cellular structure, leading to changes in the plant's leaf color and morphology indexes [66, 67]. Therefore, the

SPB material, with its ability to slowly release water and fertilizer, has great application prospects in agriculture and forestry. As shown in Fig. 7, SPB material swells upon water absorption, allowing its internal cross-linking network to unfold and slowly release urea into the soil. In the soil, urea undergoes hydrolysis to form ammonium nitrogen. This ammonium nitrogen is then absorbed by the plant root system with the assistance of microbial catalysis. A portion of nitrogen undergoes nitrification to form nitrate nitrogen, which plants can directly absorb and use. Another fraction solidifies into organic nitrogen. Additionally, higher temperatures increase urease activity, leading to the conversion of some ammonium nitrogen into ammonia, which volatilizes into the air [69, 70]. The addition of SPB material can control the release of urea, while its high water retention can reduce the soil temperature and minimize losses due to surface runoff, leaching, and volatilization, thus improving the utilization of urea. In addition, microorganisms can degrade SPB into organic nitrogen and mineralize into nitrate nitrogen and ammonium nitrogen, which plants can absorb, thus promoting plant growth. Furthermore, the addition of SPB material can improve the soil water-holding capacity, increase the water content, and enhance viscosity and swelling degree, thereby increasing soil porosity and fostering a more favorable environment for plant growth [61].

4 Conclusions

A novel slow-release fertilizer based on soy protein-based hydrogel (SPB) was synthesized, which involved graft copolymerization and semi-interpenetration of biochar, soy protein (SPI), acrylic acid (AA), and nutrients (urea) catalyzed by an initiator and cross-linker. The findings showed that the prepared SPB materials exhibited excellent expansion rates (greater than 222.3 g/g) and increased the soil water-holding capacity to more than 1.93 times its original level. Meanwhile, the loading of urea and biochar showed a significant positive correlation with the improved water retention and slow-release properties of the materials. The characterization results revealed that the introduction of biochar contributed to an increased number of cross-linking sites and enriched the network structure of SPB. As a result, with the same urea loading, SPB-3–4% exhibited an 11.7% increase in water retention on day 25 compared to SPB-3–0%, whereas both SPB materials with biochar added (SPB-2/3–4%) showed less than 44.14% urea release by day 30. In addition, the pot experiment further demonstrated that the nutrient release of SPB materials (SPB-2–4%, SPB-3–4%) with biochar addition was more in line with the growth pattern of cucumber and

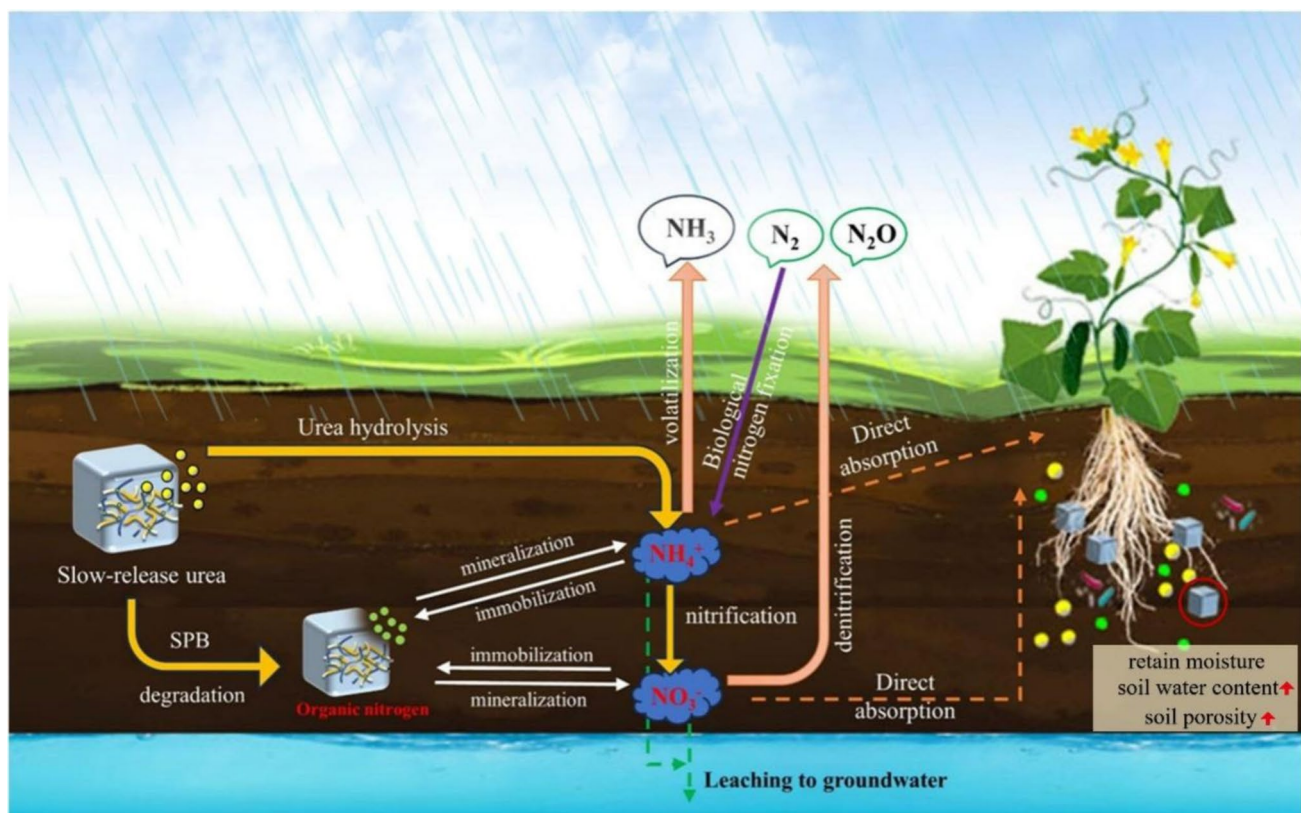


Fig. 7 Mechanism of action of SPB materials in promoting plant growth in soil

showed a better growth promotion effect. Adding 0.5% of SPB–2–4% to the soil resulted in significant enhancements in the root length, fresh weight, leaf area, and relative chlorophyll content of cucumber by 99.03%, 139.31%, 149.22%, and 109.87%, respectively. This study demonstrated that the application of SPB materials can promote plant growth, and the introduction of biochar can enhance the performance of hydrogel materials and optimize the release of water and fertilizer to better align with the plant growth patterns. Therefore, hydrogel-biochar composites offer a promising solution for soil improvement in arid and infertile areas with significant prospects in alleviating drought stress and improving fertilizer utilization.

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Data availability The data presented in this study are available on request from the corresponding authors.

Declarations

Competing interests The authors declare no competing interests.

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